SADC HUMANITARIAN AND EMERGENCY OPERATIONS CENTRE

TRAINING MANUAL

Early Warning and Monitoring Systems

SOP DATA M&A | SOP MONITORING MHD | SOP WATCH-SRA | SOP MULTI HAC

Based on SOP Volume 2, Version 3.3 — March 2026

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Version:	1.0
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SOP Reference:	Volume 2, v3.3 (SOPs 2.1–2.4)
Annexes:	22 Operational Templates (2.1-A to SR-D)
Target Audience:	EW&M Officers, Situation Room Staff, Data Analysts, NDMA Focal Points

Benchmarked: IFRC Surge Toolbox 2024 | UNDAC 8th Ed. | WMO CAP 1.2 | SADC MHEWS | ISO 22301 | OCHA AA

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1. Introduction and Purpose

This Training Manual is a companion document to the SHOC Early Warning and Monitoring Systems SOP, Version 3.3. It provides EW&M Officers, Situation Room staff, Data Analysts, and NDMA focal points with practical, SOP-by-SOP guidance on the four interconnected procedures that constitute the SHOC early warning architecture.

The four SOPs form a continuous cycle: **SOP DATA M&A (data collection and analysis) → SOP MONITORING MHD (multi-hazard monitoring) → SOP WATCH-SRA (Watch, SitRep, and Advisory issuing) → SOP MULTI HAC (anticipatory action coordination) → back to SOP DATA M&A (lessons learned feed improvements).**

⚠️ **CRITICAL MANDATE:** SHOC does not generate forecasts, alerts, or warnings. Forecasting remains the mandate of NMHS and authorised specialised institutions. SHOC consolidates, synthesises, and contextualises to produce regional coordination advisories that support anticipatory action and informed decision-making by Member States.

2. Terms and Abbreviations

The following terms are essential. A full glossary is in the SOP Ancillaries.

Abbreviation	Full Form
AAR	After-Action Review
CAP	Common Alerting Protocol
DIM	Data and Information Manager
EAP	Early Action Protocol
ED	Executive Director
EW&M	Early Warning and Monitoring
GDACS	Global Disaster Alert and Coordination System
GIS	Geographic Information System
Head EW&M	Head of Early Warning and Monitoring
Head Ops	Head of Operations
MHEWS	Multi-Hazard Early Warning System
NDMA	National Disaster Management Authority
NMHS	National Meteorological and Hydrological Services
RSMC	Regional Specialised Meteorological Centre
SHOC	SADC Humanitarian and Emergency Operations Centre
SitRep	Situation Report

3. How to Use This Manual

Each SOP section provides: SOP Purpose, Key Tasks by phase, Annex Toolkit, and Trainer's Notes. The four SOPs are interconnected — each SOP's outputs feed the next SOP's inputs.

3.1 Operational Phases

SHOC uses four descriptive operational phases that scale with risk:

Phase	EW&M Product	Staffing	Dashboard Refresh
Routine Monitoring	Watch	Duty EW&M Officer	Daily; weekly GIS refresh
Heightened Monitoring	Situation Report	EW&M Officer + EW Assistant; DIM on alert	4-hourly; hazard GIS activated
Active Response	Advisory	Full EW&M team; GIS analyst standby	Hourly; all displays active
Full Activation	Emergency Coordination	Full team rotation; surge support	Real-time; all four display walls

3.2 Cross-SOP Data Flow

SOP DATA M&A Phase 5 outputs feed SOP MONITORING MHD Phase 2 activation. SOP MONITORING MHD Phase 5 escalation triggers SOP WATCH-SRA Step 1. SOP WATCH-SRA Step 3 dissemination triggers SOP MULTI HAC Step 2. SOP MULTI HAC Step 5 lessons flow back to SOP DATA M&A Phase 1.

4. Operational Framework and Situation Room

SOP Reference: Sections 2–3

4.1 Situation Room Three-Layer Architecture

The SHOC Situation Room operates across three functional layers that trainees must understand:

Layer	Function	Key Systems
1. Data Observation & Monitoring	Receive and ingest multi-hazard feeds	GDACS, ADAM, PDC DisasterAWARE, MyDewetra, NMHS/RSMC products
2. Risk Analysis & Assessment	Process, overlay, and analyse spatial data	ArcGIS/QGIS, Power BI/Tableau dashboards, INFORM Risk Index
3. Coordination Activation	Translate intelligence into advisory content	Threshold monitoring, impact estimation, spatial attachments for NDMAs

4.2 Four Situation Room Displays

(a) Live Hazard Evolution Wall — real-time cyclone tracks, flood extent, seismic zones, epidemic spread. (b) Multi-Hazard Risk Dashboard — composite risk scores, vulnerability indicators. (c) NDMA Coordination Status Board — readiness scores, resource availability. (d) Regional Overview Map — population density, critical infrastructure, transport corridors.

4.3 Information Sourcing Hierarchy

Primary: NMHS of Member States, Regional Specialised Centres (RSMCs), SADC CSC. Continental: AMHEWAS, COMESA, EAC, IGAD. Secondary: UN agencies, GDACS, ADAM, PDC, MyDewetra. When sources conflict, primary source prevails; discrepancies logged in Data Verification Log (Annex 2.1-C).

5. SOP DATA M&A: Data Collection and Analysis

SOP Reference: Section 5 (Phases 1–6)

SOP DATA M&A governs collection, verification, standardisation, and analysis of multi-hazard data. It follows a six-phase cyclical structure.

Phase	Key Tasks	Responsible	Annex
1. Preparedness	Develop data collection plan; map hazard-region matrix; establish metadata standards; train field teams	Head EW&M, DIM, FDTL	2.1-A, 2.1-B
2. Activation	Confirm hazard event via NMHS/RSMC; seek Head Ops authorisation; mobilise field teams; coordinate access	Head EW&M, LO	2.1-C
3. Data Verification	Execute collection protocols; apply standardised instruments; ensure GPS-referenced data; document anomalies	EW&M Officer, EW Assistants	2.1-C
4. Risk Management	Assess accessibility risks using GIS layers; ensure data security per ISO 27001	Head EW&M, ICT Officer	2.1-D
5. Analysis & Reporting	Clean datasets; perform trend analysis with ArcGIS/QGIS; create dashboards; draft SitReps; submit to ED/NDMAs	Data Analysts, IMRO	2.1-D
6. Review & Recovery	After-Action Review with NDMAs; update SOP; upgrade GIS platforms and dashboard configurations	Head EW&M, ICT Officer	—

5.1 Data Quality Thresholds

Trainees must know the five quality dimensions: Accuracy (≥95% agreement with 2+ sources), Completeness (≥90% fields populated), Timeliness (≤6h latency), Consistency (0% contradictions), Validity (100% schema compliance).

5.2 Trainer's Notes

Critical exercise: Present a scenario with conflicting data from an NMHS and a secondary source (e.g. GDACS). Require trainees to apply the Information Conflict Resolution Protocol: primary source prevails, document discrepancy in Annex 2.1-C, convene technical consultation, record resolution.

6. SOP MONITORING MHD: Multi-Hazard Monitoring

SOP Reference: Section 6 (Phases 1–6 + Phase 3b)

SOP MONITORING MHD governs continuous monitoring within the Situation Room across four monitoring domains and determines when escalation to advisory products is required.

6.1 Four Monitoring Domains

Domain	Description	Display	Key GIS Layers
1. Hazard	Real-time and assessed hazard data	Live Hazard Evolution Wall	RSMC products, GDACS/ADAM/PDC/MyDewetra
2. Exposure	Population, infrastructure in hazard zones	Regional Overview Map	OCHA HDX, INFORM Risk Index, settlement data
3. Vulnerability	Socio-economic, health, institutional factors	Multi-Hazard Risk Dashboard	IPC/FEWS NET, WHO AFRO, poverty indices
4. Coping Capacity	Response capacity, preparedness, early action	NDMA Coordination Status Board	NDMA readiness assessments, stockpile locations

6.2 Key Phases

Phase 1 (Preparedness): Configure dashboards, pre-load GIS baselines, calibrate thresholds. Phase 2 (Activation): Receive trigger from SOP DATA M&A; activate hazard-specific monitoring; assign domain officers. Phase 3 (Monitoring): Track hazard evolution; update exposure/vulnerability; verify coping capacity with NDMAs; generate Watch. Phase 3b (Anticipatory Briefing): Present Situation Room intelligence to NDMAs for ground-truth verification. Phase 4 (Synthesis): Calculate composite risk scores (hazard × exposure × vulnerability/capacity); identify threshold trajectories. Phase 5 (Escalation): Update risk register; if thresholds approaching, recommend escalation to SOP WATCH-SRA. Phase 6 (Review): Deactivate; AAR; revise parameters.

6.3 Trainer's Notes

Key competency: Trainees must calculate composite risk scores by overlaying hazard, exposure, and vulnerability/coping capacity GIS layers. Set up a flood scenario and require calculation of the composite score, identification of the threshold trajectory against Annex 2.2-B, and the correct escalation recommendation.

7. SOP WATCH-SRA: Watch, SitRep and Advisory Issuing

SOP Reference: Section 7 (Steps 1–5)

SOP WATCH-SRA governs preparation, approval, dissemination, and archiving of the three SHOC EW&M products. Each product corresponds to a distinct stage of risk evolution:

Stage	Product	Focus	Conceptual Meaning	Update Cycle
Early risk	Watch	Hazard probability and trajectory	Something may happen	12–24 hours
Risk rising	Situation Report	Assessed consequences and expected impacts	This is what will likely be affected	6–24 hours
Event active	Advisory	Confirmed impacts and coordination requirements	This is what has happened; here is what we coordinate	6–24 hours

7.1 Approval Pathways

Product	Issuing Authority	Approval Time (Emergency)
Watch	EW&M Unit validation (EW&M Officer → Head EW&M)	N/A (routine product)
Situation Report	EW&M + Operations review (Head EW&M → Head Ops)	Draft ≤2h; approval ≤1h; total ≤3h
Advisory	EW&M + Operations + Leadership (Head EW&M → Head Ops → ED)	Draft ≤1h; approval ≤30min; total ≤1.5h

7.2 Steps 1–5: Advisory Workflow

Step 1: Prepare draft using SHOC templates; include GIS-derived impact maps (hazard footprint, exposed population, vulnerability hotspots, anticipatory action zones). Step 2: Review and approval per pathway above. Step 3: Disseminate via email, SHOC Portal, social media to NDMAs with spatial attachments. Step 4: Confirm receipt; escalate non-response (1h → follow-up call; 2h → Head Ops; 4h → ED; 6h → diplomatic channels). Step 5: Archive all records including spatial products in SHOC Cloud Repository by end of day.

7.3 Annex Toolkit

Annex	Title	Use
2.3-A	Advisory Message Template (CAP-aligned)	Structure advisory content
2.3-B	Advisory Approval Workflow Form	Record approval chain
2.3-C	Dissemination Channel Matrix	Primary/backup channels per recipient
2.3-D	Advisory Receipt Confirmation Log	Track receipt and escalate non-response
2.3-E	Watch Report Template	Format for Watch products
2.3-F	Situation Report Template	Format for SitRep products
2.3-G	Advisory Report Template	Format for Advisory products

7.4 Trainer's Notes

Critical exercise: Simulate a Category 3+ cyclone scenario 24 hours from landfall. Require trainees to: (a) draft an Advisory using Annex 2.3-A with GIS-derived spatial attachments, (b) execute the full approval pathway within the 1.5-hour emergency target, (c) disseminate via Annex 2.3-C channels, (d) track receipt using Annex 2.3-D and execute the escalation cascade when one NDMA fails to confirm within 1 hour.

8. SOP MULTI HAC: Anticipatory Action Coordination

SOP Reference: Section 8 (Steps 1–4)

⚠ **CRITICAL:** SHOC coordinates and advises; deployment of physical assets is the sovereign responsibility of Member States. Physical deployment is governed by the SCM SOP (Volume 8), not this SOP.

8.1 Key Steps

Step	Key Actions	Responsible	Annex
1. Pre-Disaster Preparedness	Review hazard data; establish thresholds (3-layer impact-based); pre-load GIS baselines; configure dashboard; prepare early action plans with NDMAs	Head EW&M, DIM	2.4-A
2. Anticipatory Coordination with NDMAs	Convene meetings when thresholds approach; present Situation Room spatial products; joint risk interpretation; agree proportionate actions; assign follow-up	Head EW&M	2.4-B, 2.4-C
3. Post-Disaster Review	AAR with NDMAs within 1 week; assess spatial product accuracy; compare predicted vs actual impact zones	Head EW&M	2.4-D
4. Risk Management & Continuous Improvement	Risk assessments per DIMAG; quarterly trigger review; annual calibration including Situation Room performance	Head EW&M, DIM	—

8.2 Coordination Meeting Protocol (6-Step)

(1) Convening: Head EW&M convenes when thresholds approach. (2) Evidence: Situation Room spatial products presented. (3) Shared Interpretation: NDMAs provide ground-truth. (4) Action Planning: Agree proportionate actions per Member State context. (5) Decision Recording: Record circulated within 4 hours. (6) Follow-up: Tracked on NDMA Coordination Status Board.

8.3 After-Hours Protocol

Duty Officer roster for after-hours/weekends/holidays. Duty Officer has authority to initiate Heightened Monitoring. Response times: Duty Officer immediate; Head EW&M target 30 minutes; full on-call team target 2 hours.

9. Hazard Threshold Escalation Matrix

Trainees must know the threshold levels that trigger each EW&M product:

Hazard Type	Watch	Situation Report	Advisory
Tropical Cyclone	72h landfall; <Cat 1	48h landfall; Cat 1–2	24h landfall; Cat 3+
Riverine Flood	>Alert level	>Warning level	>Danger level
Drought	SPI -1.0 to -1.5	SPI -1.5 to -2.0	SPI < -2.0
Earthquake	—	Mw 5.0–5.9	Mw ≥6.0
Epidemic	Cases >baseline	>Epidemic threshold	Outbreak declared
Food Insecurity	IPC Phase 2	IPC Phase 3	IPC Phase 4+

Full thresholds are maintained in Annex 2.2-B (Hazard Threshold Escalation Matrix). Thresholds are monitored on the Situation Room Multi-Hazard Risk Dashboard and calibrated annually plus post-activation AAR.

10. Annex Toolkit Quick Reference

All annexes grouped by SOP:

Annex	Title	SOP
2.1-A	Data Source Registry Template	DATA M&A
2.1-B	Data Collection Checklist	DATA M&A
2.1-C	Data Verification Log	DATA M&A
2.1-D	Data Quality Assurance Protocol	DATA M&A
2.2-A	CAP Advisory Message Template	MONITORING MHD
2.2-B	Advisory Approval Workflow Form	MONITORING MHD
2.2-C	Dissemination Channel Matrix	MONITORING MHD
2.2-D	Advisory Receipt Confirmation Log	MONITORING MHD
2.3-A	Advisory Message Template (CAP-aligned)	WATCH-SRA
2.3-B	Hazard Threshold Escalation Matrix	WATCH-SRA
2.3-C	Shift Handover Log	WATCH-SRA
2.3-D	Hazard Evolution Assessment Report	WATCH-SRA
2.3-E/F/G	Watch, SitRep, and Advisory Report Templates	WATCH-SRA
2.4-A	Anticipatory Action Trigger Protocol	MULTI HAC
2.4-B	EAP Activation Checklist	MULTI HAC
2.4-C	Resource Pre-Positioning Request Form	MULTI HAC
2.4-D	After-Action Review Template	MULTI HAC
SR-A to SR-D	Situation Room Operations Annexes (Display Config, GIS Standards, Refresh Protocol, Access Control)	Sit Room

11. Decision Authority and Escalation

11.1 Decision Authority Matrix

EW&M Product	Trigger	Authority (From)	Authority (To)
Watch	Watch recommends SitRep be developed	EW&M Officer	Head EW&M
Situation Report	SitRep recommends Advisory be developed	Head EW&M	Head Ops
Advisory	Threshold breach confirmed; NDMA coordination required	Head EW&M	Head Ops / ED
Coordination Meeting	Multi-country coordination required; AA triggers approaching	Head EW&M	ED

11.2 Receipt Escalation Cascade

1 hour: Follow-up call to NDMA primary contact (Liaison Officer). 2 hours: Head Ops notified; alternative channels activated. 4 hours: ED briefed; SADC Secretariat diplomatic channels considered. 6 hours: Documented in Annex 2.3-D; flagged for AAR.

12. Risk Analysis and Performance Indicators

Key performance indicators trainees must know:

Indicator	Target	Frequency
Advisory draft-to-dissemination (Emergency)	≤1.5 hours	Per activation
SitRep draft-to-dissemination (Emergency)	≤3 hours	Per activation
Data quality — Accuracy	≥95%	Per collection cycle
Data quality — Timeliness	≤6 hours latency	Per collection cycle
NDMA receipt confirmation	100% within 1 hour	Per dissemination
Situation Room dashboard refresh (Active)	Hourly	During Active Response
Composite risk score generation	Per monitoring cycle	Continuous
After-Action Review completion	≤1 week post-event	Per activation
Threshold calibration review	Annual + post-AAR	Annual
After-hours response time (Head EW&M)	≤30 minutes	Per after-hours event

13. Assessment and Competency Verification

Competency in SHOC Early Warning and Monitoring is assessed through practical exercises aligned to the four-SOP structure.

Competency	Assessment Method	Minimum Standard
Information Sourcing Hierarchy	Written test: rank sources correctly	Correct hierarchy; conflict resolution protocol
Data collection and verification	Complete Annexes 2.1-A/B/C from scenario data	All fields complete; quality thresholds met
Composite risk score calculation	GIS exercise: overlay hazard/exposure/vulnerability	Correct score; threshold trajectory identified
Four monitoring domains	Scenario: assign monitoring and produce Watch	All 4 domains covered; Watch format correct
Watch → SitRep → Advisory escalation	Simulate escalation chain with time pressure	Correct product at each stage; approval pathways correct
Advisory drafting with GIS products	Produce Advisory with spatial annexes	CAP-aligned template; 4 spatial annexes attached
Receipt escalation cascade	Simulate NDMA non-response	Correct escalation at 1h/2h/4h/6h
Anticipatory coordination meeting	Role-play: lead 6-step meeting protocol	All 6 steps completed; decision record within 4h
Threshold application by hazard	Written test: apply thresholds to 6 hazard types	Correct product for each scenario
After-hours activation	Scenario: activate from Duty Officer position	Heightened Monitoring initiated; Head EW&M within 30min

Certification requirement: EW&M Officers and Situation Room staff must demonstrate competency across all ten areas before assuming operational duty. Quarterly readiness drills and seasonal preparedness exercises provide the primary assessment vehicle. Results are recorded in the SHOC EW&M competency register and reviewed by the Head of Early Warning and Monitoring.

— END OF TRAINING MANUAL —